MOST ASKED SQL INTERVIEW QUESTION

SEEN IN INTERVIEWS AT

Amazon | JP Morgan | Infosys | TCS

FIND THE LAST SUNDAY

FIND THE NEXT SUNDAY

SWIPE FOR PATTERNS ➡️

THE LOGIC

THE UNIVERSAL PATTERN

PATTERN 1: LAST (BACKWARD)

```
DATE_SUB( ... (Current - Target + 7) % 7 )
```

PATTERN 2: NEXT (FORWARD)

```
DATE_ADD( ... (Target - Current + 7) % 7 )
```

BEGINNER'S NOTE

SQL calculates days using numbers.
Usually, Sunday = 1 or Sunday = 6.

`+ 7 % 7` : The "Modulo" Trick.
It wraps the calculation within a 7-day week so you don't get negative days.

Follow for SQL Patterns

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REFERENCE

KNOW YOUR NUMBERS

SYSTEM 1: SUNDAY START

Used by MySQL (DAYOFWEEK) and SQL Server (Default).

SUN	MON	TUE	WED	THU	FRI	SAT
1	2	3	4	5	6	7

SYSTEM 2: MONDAY START

Used by MySQL (WEEKDAY) and SQL Server (SET DATEFIRST 1).

MON	TUE	WED	THU	FRI	SAT	SUN
0	1	2	3	4	5	6

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SUN START

SUNDAY START (1-7)

MYSQL

⬅️ Last Sunday — DAYOFWEEK

```mysql
-- Sun = 1
SELECT DATE_SUB(now(), INTERVAL (
  (DAYOFWEEK(now()) - 1 + 7) % 7
) DAY);
```

➡️ Next Sunday — DAYOFWEEK

```mysql
-- Sun = 1
SELECT DATE_ADD(now(), INTERVAL (
  (1 - DAYOFWEEK(now()) + 7) % 7
) DAY);
```

SQL SERVER

⬅️ Last Sunday — DATEPART

```sql server
-- Sun = 1
SELECT DATEADD(day, -(
  (DATEPART(dw, GETDATE()) - 1 + 7) % 7
), GETDATE());
```

➡️ Next Sunday — DATEPART

```sql server
-- Sun = 1
SELECT DATEADD(day, (
  (1 - DATEPART(dw, GETDATE()) + 7) % 7
), GETDATE());
```

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MON START

MONDAY START (0-6)

MYSQL

⬅️ Last Sunday — WEEKDAY

```
-- Sun = 6
SELECT DATE_SUB(now(), INTERVAL (
  (WEEKDAY(now()) - 6 + 7) % 7
) DAY);
```

➡️ Next Sunday — WEEKDAY

```
-- Sun = 6
SELECT DATE_ADD(now(), INTERVAL (
  (6 - WEEKDAY(now()) + 7) % 7
) DAY);
```

SQL SERVER

⬅️ Last Sunday — DATEFIRST 1

```
SET DATEFIRST 1; -- Sun = 7
SELECT DATEADD(day, -(
  (DATEPART(dw, GETDATE()) - 7 + 7) % 7
), GETDATE());
```

➡️ Next Sunday — DATEFIRST 1

```
SET DATEFIRST 1; -- Sun = 7
SELECT DATEADD(day, (
  (7 - DATEPART(dw, GETDATE()) + 7) % 7
), GETDATE());
```

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SUMMARY

MySQL Strategy

Use **WEEKDAY (0-6)** for Mon=0.

Use **DAYOFWEEK (1-7)** for Sun=1.

SQL Server Strategy

Function is always **DATEPART**.

Use **SET DATEFIRST** to change.

+7 % 7

Ensures the calculation wraps around the week correctly (Modulo).

SAVE THIS CHEATSHEET 💾

Follow for SQL Patterns

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